Supplementary file number 4 (S4) : Detailed summary of the results of individual studies

Study (Author, Year, Country)	intervention	staff	parents	Outcomes	Tools	Results		Positive/Negative Outcome	Pre Scores	T1	T2	Remarks
Burns-Nader ES, 2011, USA	1)a medical play group, where children were given a doll and authentic medical equipment, 2) a medical information video group, in which children were shown a taped medical play session, 3) a typical play group (i.e., played Connect Four), or 4) a control group whose participants viewed a video on safari life.	1. nurses completed behaviour assesment scale	parents' self-reported level of worry about the doctor's visit positively correlated with their reports of the children's level of worry about the doctor's visit and the children's baseline pulse.						MEAN SCORES			Relationship between variables
				Anxiety level	The Child Drawing: Hospital (Higher score indicates higher anxiety) Heart rate	Not a significant effect of type of activity on the children's level of anxiety, as depicted in their drawings		NO EFFECT				
				Level of fear	Fear Scale (5-points likert scale) (Higher score indicates higher level of fear)	1. Increase in fear for children in the medical play group, from baseline (1.44) to post-activity (1.67), suggesting that children in the medical play activity became more fearful after medical play		NEGATIVE (MEDICAL PLAY)	1.44	1.67 (post-activity)		
						2. Increase in fear for children in the medical play activity from baseline (1.44) to after the doctor's visit (1.94), suggesting that children in the medical play activity were more fearful after the doctor's visit compared to the baseline		NEGATIVE (MEDICAL PLAY)	1.44		1.94 (after doctor's visit)	
						3. Decrease in fear for children who watched the medical information video from baseline (2.28) to post activity (1.31) and from baseline (2.28) to after the doctor's visit (1.50), suggesting that the medial information video was effective in reducing children's fear after the activity and after the doctor's visit.		POSITIVE (MEDICAL VIDEO)	2.28	1.31 (post-activity)		
								POSITIVE (MEDICAL VIDEO)	2.28		1.5 (after doctor's visit)	
				Children's distress behaviours	Behaviour Observation Scale (BOS) (Higher score indicates less alert) Children's Hospital of Eastern Ontario Pain Scale (CHEOPS) (Higher score indicates more difficulty)	Children in the typical play group improved in their state, or alertness, from baseline (1.94) to post-activity (1.72)		POSITIVE (TYPICAL PLAY)	1.94	1.72 (post-activity)		
						The video control group shifted from an active alert state at baseline (1.28) to a more inactive alert state (1.72) after watching the safari video. The decrease in alertness also occurred from baseline (1.28) to after the doctor's visit (1.67), suggesting that they were most alert before the activity		NEGATIVE (NON-MEDICAL VIDEO)	1.28	1.72 (post-activity)		
								NEGATIVE (NON-MEDICAL VIDEO)	1.28		1.67 (after doctor's visit)	
						The medical play group has higher mean than the rest of the activity groups on nurses's report on overall difficulty suggests that children in the medical play group were more difficult than any other group while the nurse was performing triage procedures		NEGATIVE (MEDICAL PLAY)				
Brown JC, 2012, USA	1. The medical play collage consisted of medical materials, such as band aids, tongue depressors, gauze, medical tape and small syringes without needles. 2. The control group received no intervention and continued doing whatever typical activity they were already doing	1. researchers	no significant assoiation between: parents talking about doctor visit to the children and parental pereception about worry of child between control and intervention group	Children's behaviour and feelings	Behaviour Observation Scale (BOS) (Higher score indicates more alert) Faces Scale (4-points likert scale) (Higher score indicates more positive mood)	The intervention group showed significant differences between time points pre and post one and post one and post two. These differences were seen between pre and post one state (Z = -2.00, p = .046), pre and post one activity (Z = -2.83, p = .005), post one and post two activity (Z = -2.45, p = 0.14), pre and post one vocalization (Z = -2.83, p = .005), and post one and post two vocalization (Z = -3.05, p = .002). This indicates that after the intervention, the participants became more alert, and had more activity and more vocalizations. After the intervention, levels of activity and vocalization went back down.			FREQUENCIES			
							State	POSITIVE	Inactive alert: 5 (Pre) Active alert: 10 (Pre)	Inactive alet: 1 (Post-one) Active alert: 14 (Post-one)		Pre - before intervention Post one - right after intervention Post two - after doctor's visit
							Activity	POSITIVE	Moderate: 11 (Pre) High: 2 (Pre)	Moderate: 5 (Post-One) High: 9 (Post-One)	Moderate: 11 (Post-Two) High: 3 (Post-Two)	
							Vocalisation	POSITIVE	Low: 4 (Pre) High: 4 (Pre)	Low: 1 (Post-one) High: 9 (Post-one)	Low: 7 (Post-two) High: 4 (Post-two)	
Ottenheim M, 2018, Netherlands	On the day of the visit to the Teddy Bear Hospital every child brings their own teddy bear with an undetermined medical problem, again playing the role of the parent. With their teddy bear, the child will go through several rooms of the hospital mimicking a real hospital visit: the waiting room, the consulting room, the x- ray room, the operation room or the plaster room and finally the pharmacy. In these room the child and their teddy bear will be helped by several different doctors.ols were supplied with educational materials around the theme "hospital", to prepare for the visit. The package included picture books, a stuffed animal, songs, colouring activities and explanatory posters.	These 'doctors' are actually student medical doctors playing the role of doctor	NA						MEAN SCORES			
				1. Children's attitude towards visiting a hospital 2. Children's attitude towards doctors 3. Children's attitude towards falling ill 4. Children's attitude towards hospitals	Seven-point likert scale (Higher score indicates more positive attitude)	Significant difference in anxiety level for doctors between the three measure moments. Gender, if children visited a hospital before or the interaction between the covariates did not differ significantly. The Bonferroni pairwise comparison showed significant differences between T2 (5.34) and T0 (5.00) (p=0.002); T2 (5.34) and T1 (5.00) (p=0.018) but not between T0 (5.00) and T1 (5.00) (p=1.00).		POSITIVE	5.00	5.00	5.34	T0: before the lesson with the materials provided by the IFMSANL. T1: after the lesson with the materials provided by the IFMSA-NL but before the actual visit to the Teddy Bear Hospital. T2: after the visit to the Teddy Bear Hospital.
						Significant difference in anxiety level for falling ill between the three measure moments. Gender, if children visited a hospital before or the interaction between the covariates did not differ significantly. The Bonferroni pairwise comparison showed significant differences between T0 (2.72) and T1 (3.23) (p=0.000); T0 (2.72) and T2 (3.25) (p=0.000) but not between T1 (3.23) and T2 (3.25) (p=1.00).		POSITIVE	2.72	3.23	3.25	
						Gender had a significant effect on anxiety levels for going to the hospital ill with girls reporting on average higher levels of anxiety (low on the Likert scale) compared to boys.			Boy: 4.42 Girl: 4.40	Boy: 4.83 Girl: 4.14	Boy: 4.63 Girl: 4.24	
Ong L, 2018, Singapore	Teddy Bear Hospital is a simulated environment of a patient's visit to hospital. At each kindergarten or within each class at the primary school, the children were divided into 4 groups to participate in 4 concurrent stations in turns, before coming together for the final station. There are 4 stations depicting various clinical areas in a hospital, namely the Family Medicine Clinic, the Asthma clinic, the Orthopaedics Clinic and the Operation Theatre. Participants will learn about the various specialties depicted in each station through role-play as young doctors taking care of their teddy bears or stuffed toys. Each station will last 6 to 8 minutes. The entire Teddy Bear Hospital session (rotation through all 5 stations) takes approximately 45minutes to 1 hour.	The programme is fully run by the medical student volunteers.	Parents were not present during the TBH sessions						FREQUENCY PERCENTAGE			
				Childhood feelings towards healthcare services & willingness of health-seeking behaviour:	5-point likert scale (Higher score indicates more positive mood)	Children's feeling of happiness towards visiting the doctor increased from 82.5% to 94.4% (p-value <0.001) with the median changing from 2 to 1.		POSITIVE	82.50%	94.40%		
						Children who felt happy to visit the hospital increased from 70.2% to 73.4% (p-value 0.035) when measured on the same 5-point scale.		POSITIVE	70.20%	73.40%		
				Children's healthcare knowledge	3 health-care related questions	Children who answered all three knowledge-related questions correctly increased from 81 (32.1%) before attending TBH to 185 (73.4%) after attending TBH, with the median value improving from 2 to 3 (p-value <0.001)		POSITIVE	32.10%	73.40%		
Dalley JS, 2016, Canada	45-minute tour children spent most of their time learning about general medical equipment in the main exam room (e.g., needles, stethoscope, and otoscope); they also briefly visited X-ray machines in the radiology room and physical therapy in the rehabilitation room. Children were taught coping strategies throughout the entire tour, including an initial introduction to the coping strategies at the beginning of the tour, followed by reminders and further rehearsal during each scheduled room of the tour. The tour script (see the Appendix) was based on the pediatric medical communication literature. The script utilized concrete language and short statements regarding the reason for medical procedures, the medical equipment used, and the sensations children would feel, with analogies included when applicable (e.g., using a stethoscope makes listening to your breath easier because it is louder, like turning up the volume on a TV). There is some consensus that children learn information best through demonstration [24] and therefore medical procedures were described and demonstrated by tour leaders on each child's stuffed animal	The tour was led by research assistants from the second author's Pediatric Pain, Health and Communication Lab and veterinary students from the Primary Healthcare Centre.	NA						MEAN SCORES			
				Level of fear	The Children's Fear Scale (5-points likert scale) (Higher score indicates higher level of fear)	Children's fear ratings did not signifcantly change from pre to post intervention						Time 1 - immediately before the tour Time 2 - immediately after the tour (45 minutes later)
				Level of expected pain	The Faces Pain Scale - Revised (0-10 metric) (Higher score indicates higher level of expected pain)	Mean expected pain ratings decreased signifcantly from Time 1 (M = 2.00, SD = 1.27) to post intervention at Time 2 (M = 1.64, SD = 1.35)		POSITIVE	2.00	1.64		
				Coping strategies	14-item coding system (Higher score indicates more coping strategies adapted)	Children reported a higher frequency of intervention-taught coping strategies at Time 2 (M = 0.90, SD = 0.72) versus Time 1 (M = 0.70, SD = 0.76), t(70) = −2.17, p = .034, and r = −.06, representing a signifcant, albeit small effect		POSITIVE	0.7	0.9		
Pontes JE, 2015, Brazil	Instructional therapeutic play sessions were carried out individually, using a doll, disposable syringes, needles, cotton, and bandages for venal puncture. The researcher showed the procedure in the doll and asked the child to participate and repeat the procedure at the end of the role play.	reseacher	NA	1. Children's body movement during procedure 2. Children's verbal expression during procedure 3. Children's emotional expression during procedure	Child's reaction checklist	Most reactions related to a greater acceptance of the procedure, such as staying still, smiling, collaborating, and being at ease, were more frequent in the experimental group. Fisher's exact test showed that the differences were significant for standing still (p=0.006) and collaborating spontaneously (p=0.015), with significance level at 5%. Some of the reactions that show less acceptance of the procedure were not observed in the experimental group, such as hitting, kicking, pushing, biting, pulling the hand of the professional, clinging to their parents, asking for cuddling and showing muscle stiffness, pallor, and perspiration. Fisher's test showed that differences were significant for pushing (p=0.005), clinging to their parents (p=0.000), asking for cuddling (p=0.024), and muscle stiffness (p=0.002).		POSITIVE				
Leonhardt C, 2014, Germany	The visit to the TBH starts with registration when the child with his or her stuffed animal gives his or her name and describes the illness of the soft toy. Then the child can play in a waiting room until a contact person accompanies the child to his or her teddy doctor. Each teddy doctor wears a lab coat and has the usual medical office equipment (e.g. stethoscope, clinical thermometer, syringes and adhesive tapes). In an examination room that is quite realistic, the teddy doctor carries out the physical examination, depending on the individual illness of the soft toy. The children are encouraged to ask questions and take part in the examination. After the consultation, teddy may be referred to the teddy surgeons, to a dentist, to a place with ultrasound, scales or X-ray simulation (camera connected with a computer) or to the operating theatre. The teddy doctors converse with the children in an easy and responsive manner, depending on the child's verbal capability and emotional state. Finally, the children are given a prescription and go to the TBH pharmacy, where they are provided with tea, warming pans, ice bags, adhesive tapes or bandages from students from the School of Pharmacy. An important aspect of prescription for each child is the 'snuggle therapy' that is recommended for each individual case. The child has the chance of looking into an ambulance and finally goes back to the waiting area, where the nursery teacher or parents wait.	'teddy docs' (graduate medical students). The organizing committee is made up of approximately 20 students.	NA						MEAN SCORES			
				Children's medical knowledge	Children's Medical Knowledge Scale (ChildMedKnow) (Higher score indicates better knowledge)	Children who visited the TBH had a significantly better knowledge of their body, health and disease (ChildMedKnow at time point 2) than the children from the control group (F(1, 118) = 141.07, p < 0.001). The effect size was large (partial η2 = 0.55). There was an influence of the baseline score as well (F(1, 118) = 213.55, p < 0.001, partial η2 = 0.65).		POSITIVE				Time 1 - during the week prior to the intervention Time 2 - 1 week after visiting the TBH
						Children who visited the TBH improved their knowledge of the different organs of the body significantly (F(1, 121) = 79.43, p < 0.001, partial η2 = 0.41). Children from the intervention group knew the organs of their body better after visiting the TBH (M = 4.5, SD = 1.1) than children in the control group (M = 3.1, SD = 1.2). Baseline M = 1.93		POSITIVE	1.93	Intervenion: 4.5 Control: 3.1		
						The ANCOVA also revealed here a significant baseline score as covariate (F(1, 121) = 111.05, p < 0.001, partial η2 = .48).						
Burns-Nader S, 2013, USA	Child participants were randomly assigned to 1 of 4 groups: (a) a medical play group, where children were given a doll and authentic medical materials to play with (ie, stethoscope, otoscope, tongue depressor, etc); (b) a medical information video group, in which children were shown a taped medical play session; (c) a typical play group, where children played a board game; or (d) a control group whose participants viewed a video on safari life.	ressearcher	NA						MEAN SCORES			
				Children's mood	Mood Scale (5-poins likert scale) (Higher score indicates less positive mood)	A significant interaction was found between time and type of activity on children's self-reported mood, F(5.92,122.31) = 3.43, P < .01. Paired samples t tests were conducted for post hoc comparisons within groups and revealed lower scores in mood (ie, shifting from the direction of very happy to more in the direction of happy) for children in the medical play group, from baseline to postactivity and from baseline to after the doctor's visit, suggesting the medical play activity lowered the children's mood after the activity (1.67) and after the doctor's visit (1.94). Baseline mean = 1.44			1.44	1.67 (Post-intervention)		
								NEGATIVE (MEDICAL PLAY)	1.44		1.94 (Post-doctor's visit)	
						Higher scores in mood (ie, shifting from the direction of happy to more in the direction of very happy) for children who watched the medical information video from baseline to post activity and from baseline to after the doctor's visit, suggesting that the medical information video improved children's mood after the activity (1.31) and after the doctor's visit (1.5). Baseline mean = 2.28			2.28	1.31 (Post-intervention)		
								POSITIVE (MEDICAL VIDEO)	2.28		1.5 (Post-doctor's visit)	
				Children's behaviour during procedure	Nurse's rating on Children's behaviours (Higher score indicates more distress)	Significant effect for children's leg behaviors, F(3) = 2.85, P < .05, and the children's overall difficulty during the triage procedures, F(3) = 5.63, P < .01. Post hoc results for leg behaviors revealed significant difference between the medical play group and the nonmedical information video control, with the medical play group having a higher mean, suggesting that they were more likely to display distress behaviors with their legs than the nonmedical information video group during triage.		NEGATIVE (MEDICAL PLAY)				
						The medical play group was also significantly different than the medical information video, typical play, and nonmedical information video groups on the nurses' overall rating of how difficult the children were during the triage procedures. The medical play group had a higher mean than the rest of the activity groups. This would suggest that children in the medical play group were more difficult than any other group while the nurse was performing triage procedures.		NEGATIVE (MEDICAL PLAY)				
									FREQUENCY			
Victorine A, 2002, UK	The child will be taken through to Hygiene Talk where they are taught about cleanliness, hygiene and hand-washing. After that, they will then attend the 'Talk of Fears' – where in a group of 5-6 children, they talk about their fears and group them in a pyramid. Working from the bottom of the pyramid, we talk around the fears using methods similar to systemic desensitization with relaxation and breathing exercises in between exposure. Children are encourage to think about their fears, then relax, imagining themselves there, then relax, seeing a picture of what the child fears, relax and eventually going through it and being able to talk about it and feel less scared about it. From there, the child is then taken through to 'TBH Consultation' where the role-play begins. Teddy doctor will take a history from poorly teddy and teddy's parent (the child), then decide to perform some examination – inspection, palpation, percussion, auscultation, taking teddy's temperature, blood pressure, x-ray and CAT scan. Depending on the child's capability, the consultation can be modified to involve the parents – i.e. child taking mum or dad to see teddy doctor, role-play swap where teddy doctor becomes the patient and the child becomes the doctor to allow them hands-on experiences with medical equipments etc. After the consultation, teddy may be referred to the teddy	Teddy doctors were recruited from the medical school via MedSoc.	NA	1. Children's attitude towards doctor 2. Children's attitude towards hospital 3. Children's attitude towards TBH	7-point likert scale (Higher score indicates more positive mood)	1. Of the 57 children, 13 of them gave a score of 6 on the Likert scales when asked if they enjoyed Teddy Bear Hospital while 44 of them gave a score of 7. 2. When asked if Teddy enjoyed Teddy Bear Hospital, 4 of the children gave a score of 5, 19 children gave a score of 6 and the balance 34 gave a score of 7. 3. Of the 57 children, when asked if Teddy Bear Hospital have helped them to reduce their fears, 55 of them answered yes and 2 not. Of the two that said not, both expressed no fear of doctors and hospital at all from the beginning. 4. All the children answered yes when asked if they would attend Teddy Bear Hospital again, if teddy would attend Teddy Bear Hospital again, if the children will invite their friends to Teddy Bear Hospital and if teddy will bring teddy's friend to Teddy Bear Hospital.		POSITIVE	Likert score 7 (most happy and comforable) Doctor: 0 Hospital: 0	Likert score 7 (most happy and comforable) Doctor: 32 Hospital: 32		Pre and post TBH graph
									MEAN SCORES			
Bloch YH, 2008, Israel	Children were invited to attend the simulated hospital where they would act as parents of their own teddy bears. Beforehand they were asked to think of a disease for their teddy bear. This activity was facilitated by the kindergarten teacher, but the choice of the illness or medical condition was the child's alone. All of the "teddy bear hospital" activities took place in the Soroka University Medical Center's main courtyard. This allowed proximity to the hospital facilities, but in a pleasant atmosphere and without exposing the children to the possible risks of the wards, such as infectious diseases, other children's pain, etc. The acting physicians conducted a structured protocol of medical procedures including history taking, physical examinations, and various laboratory and diagnostic tests. The history was taken in a flexible and responsive manner, according to the child's own verbal capability and using his or her own	Twenty medical students at the Ben-Gurion University's Faculty of Health Sciences played the role of the physicians	NA	Children's anxiety about hospitalisation	One-item visual analog scale of anxiety about hospitalization (5 facial expressions) (Higher score indicates higher level of anxiety)	A repeated measures ANOVA revealed that time signifcantly interacted with group [F(1,89) = 14.34, P < 0.001]. While baseline levels of anxiety were not different between groups [t(89) = 0.4, NS], children in the teddy bear hospital group reported signifcantly lower levels of anxiety than the control group at follow-up		POSITIVE	TBH: 6.5 Control: 7.0	TBH: 5.0 Control: 8.0		Pre and post TBH graph

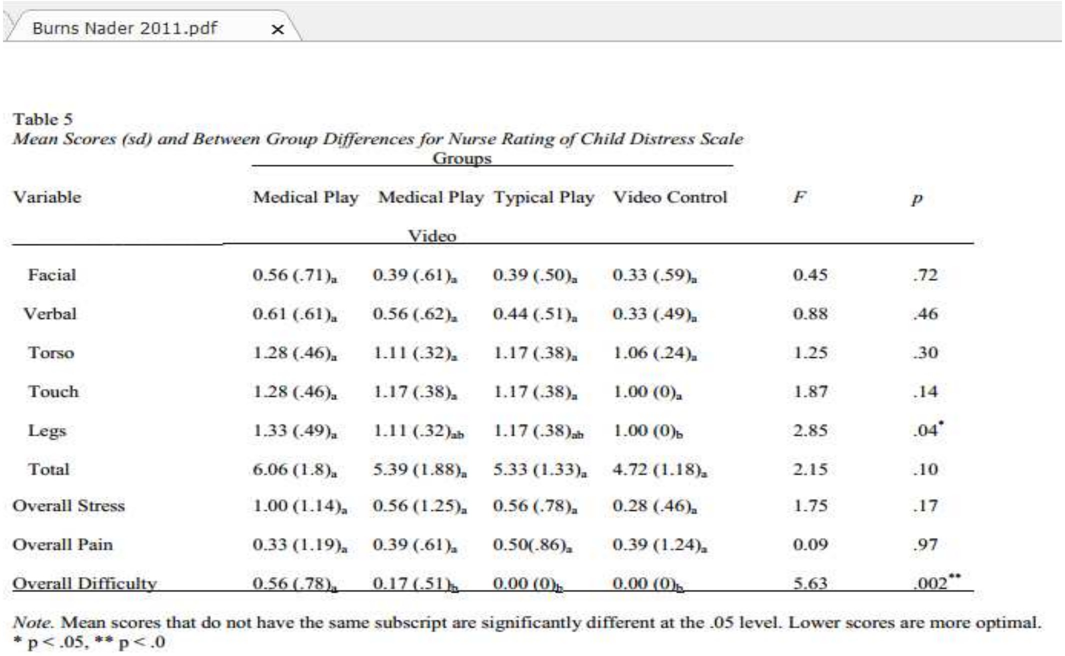
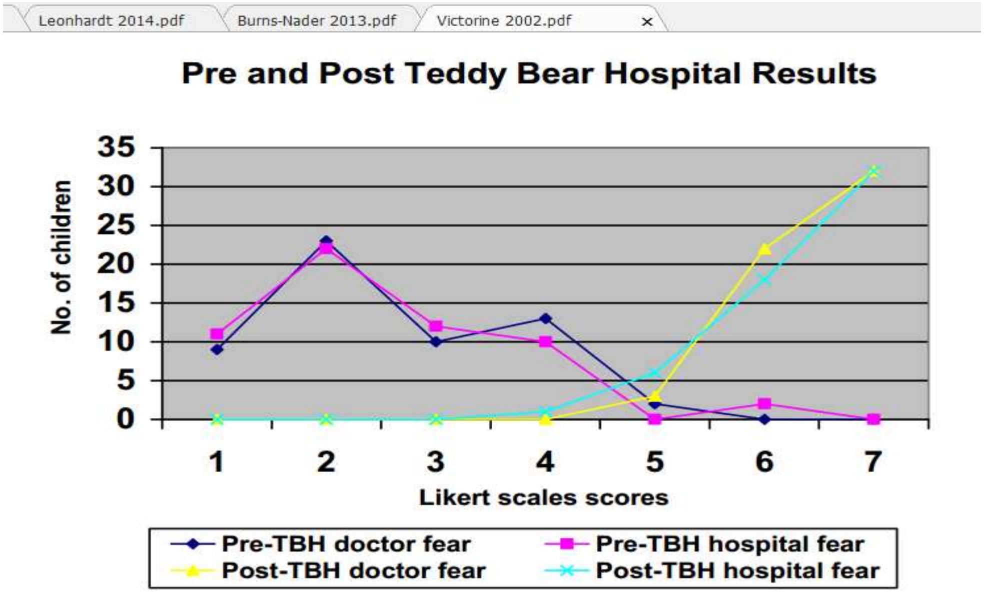


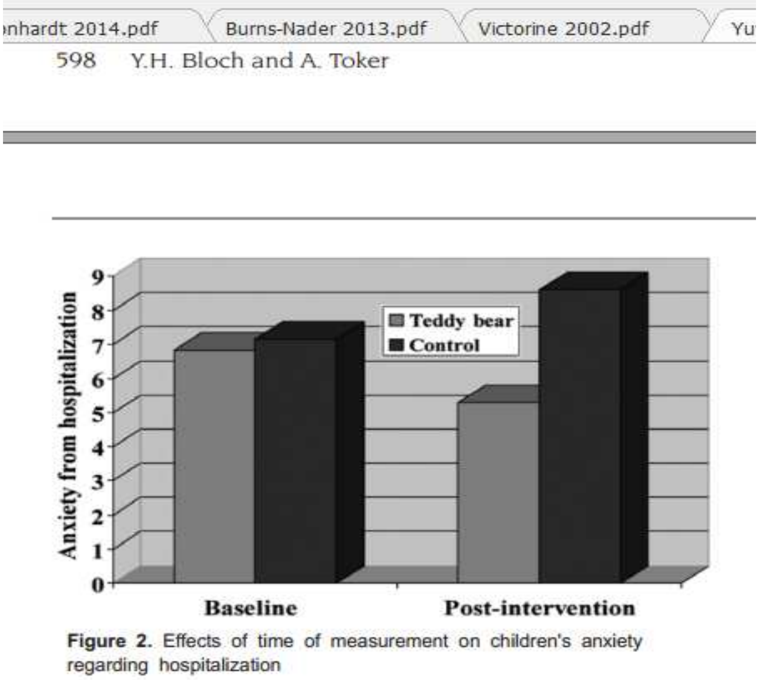
Table 5

Mean Scores (sd) and Between Group Differences for Nurse Rating of Child Distress Scale

	Groups					
Variable	Medical Play	Medical Play Video	Typical Play	Video Control	*F*	*p*
Facial	0.56 (.71)$_a$	0.39 (.61)$_a$	0.39 (.50)$_a$	0.33 (.59)$_a$	0.45	.72
Verbal	0.61 (.61)$_a$	0.56 (.62)$_a$	0.44 (.51)$_a$	0.33 (.49)$_a$	0.88	.46
Torso	1.28 (.46)$_a$	1.11 (.32)$_a$	1.17 (.38)$_a$	1.06 (.24)$_a$	1.25	.30
Touch	1.28 (.46)$_a$	1.17 (.38)$_a$	1.17 (.38)$_a$	1.00 (0)$_a$	1.87	.14
Legs	1.33 (.49)$_a$	1.11 (.32)$_{ab}$	1.17 (.38)$_{ab}$	1.00 (0)$_b$	2.85	.04*
Total	6.06 (1.8)$_a$	5.39 (1.88)$_a$	5.33 (1.33)$_a$	4.72 (1.18)$_a$	2.15	.10
Overall Stress	1.00 (1.14)$_a$	0.56 (1.25)$_a$	0.56 (.78)$_a$	0.28 (.46)$_a$	1.75	.17
Overall Pain	0.33 (1.19)$_a$	0.39 (.61)$_a$	0.50(.86)$_a$	0.39 (1.24)$_a$	0.09	.97
Overall Difficulty	0.56 (.78)$_a$	0.17 (.51)$_b$	0.00 (0)$_b$	0.00 (0)$_b$	5.63	.002**

Note. Mean scores that do not have the same subscript are significantly different at the .05 level. Lower scores are more optimal.
* $p < .05$, ** $p < .0$


Leonhardt 2014.pdf
Burns-Nader 2013.pdf
Victorine 2002.pdf
×
Pre and Post Teddy Bear Hospital Results
No. of children
35
30
25
20
15
10
5
0
1
2
3
4
5
6
7
Likert scales scores
Pre-TBH doctor fear
Pre-TBH hospital fear
Post-TBH doctor fear
Post-TBH hospital fear



598 Y.H. Bloch and A. Toker

Figure 2. Effects of time of measurement on children's anxiety regarding hospitalization